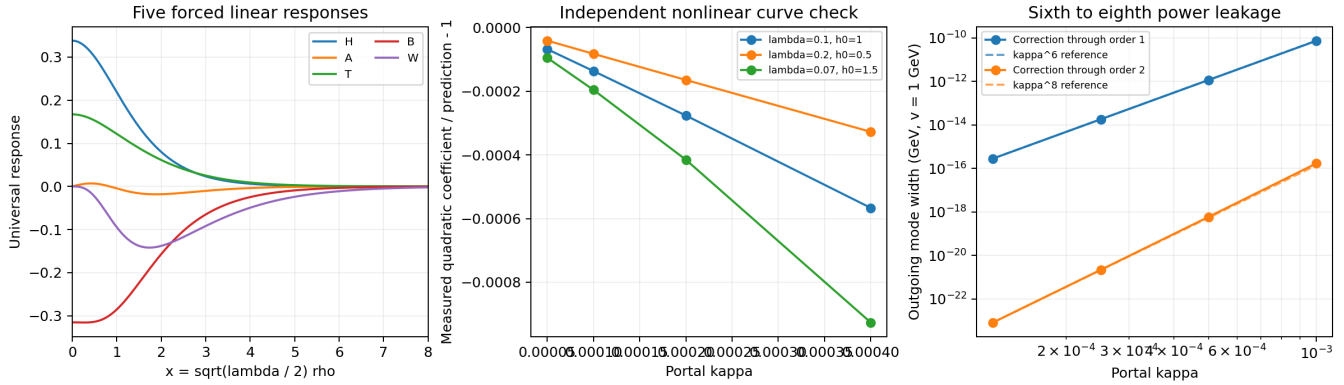


Polynomial wall responses and the formal radiation-cancellation hierarchy

The retuned wall calculation now predicts the curvature of its radiation-node condition from five forced linear responses. It no longer needs to infer that coefficient from the nonlinear candidate curve. In the decoupled-singlet case, the source quartic and Higgs vacuum ratio enter the result through an explicit universal scaling law. Independent coupled background/eigenmode calculations check that law at three parameter choices. Independent outgoing-pole calculations change the observed small-portal leakage from sixth-power to eighth-power behavior when the new quadratic correction is applied. The deeper mathematical extension is an order-by-order Fredholm construction: parity leaves one real open-channel solvability condition, and the next Higgs-quartic coefficient enters it with a nonzero overlap. This constructs a formal localized continuation, without establishing convergence or infinite-domain existence.



The declared action and normalization. Work in $\rho=v z$, with u the dimensionless odd source, h the dimensionless even Higgs radius, fixed $h_0=v_H/v>0$, source quartic $\lambda>0$, and portal κ . Set the Gaussian mediator coupling to zero, so that the spectator singlet decouples exactly. The square-completed potential, divided by v to the fourth, is

$$V = \frac{\lambda}{4}(u^2 - 1)^2 + \frac{\lambda_H}{4} \left[h^2 - h_0^2 + \frac{\kappa}{\lambda_H}(u^2 - 1) \right]^2.$$

At $\kappa=0$, put $k=\sqrt{\lambda/2}$, $x=k \rho$, $u=\tanh(x)$, $s=\text{sech}(x)$, and $p=s \tanh(x)$. The radiation-node Higgs quartic is $\lambda_H h_0=k^2/h_0^2$. The source shape energy is $3k^2$; its dimensionless odd mode is p/k , normalized to source derivative one at $\rho=0$. The open Higgs wave number is k , and the first induced Higgs mode is $\kappa h_0 s/k^3$. Both modes decay even though the Higgs channel is energetically open. The previous release derived the first threshold correction C approximately -1.36468352107739 . The present calculation keeps the full square completion, including the κ^2/λ_H source terms, while expanding the background and mode consistently.

The five universal responses. Primes below denote derivatives with respect to x . H is the first Higgs background response; A is the second source background response; T is the second Higgs background response; B is the second Higgs-mode response; W is the second source-mode response. Their scaling definitions are

$$h = h_0 + \frac{\kappa h_0}{k^2} H + \frac{\kappa^2 h_0^2}{k^4} T + \dots, \quad u = \tanh x + \frac{\kappa^2 h_0^2}{k^4} A + \dots.$$

$$\psi = \frac{p}{k} + \frac{\kappa^2 h_0^2}{k^5} W + \dots, \quad \chi = \frac{\kappa h_0}{k^3} s + \frac{\kappa^2 h_0}{k^5} B + \dots.$$

The response H satisfies $H''=2Hs^2$, with even parity and decay. The source response A satisfies $A''=(4-6s^2)A+\tanh(x)(2H-s^2)$, with odd parity and decay. The second static Higgs response satisfies $T''=2T+3Hs^2+CH-Hs^2$. The second open mode satisfies $B''+B=s[(8-2s^2)H+C-s^2]$. Requiring B to be even and decaying fixes C . This is the same cosine solvability condition as the earlier Green-response calculation, now appearing inside the coupled response hierarchy. No outgoing width enters these equations or the determination of C .

The source-mode Fredholm condition. The odd shape operator has kernel p . Its order-two response satisfies

$$W'' = (1 - 6s^2)W + (12uA + 2H + 3u^2 - 1 - e)p + 2us, \quad u = \tanh x.$$

Odd parity requires $W(0)=0$. The mode normalization requires $W\text{-prime}(0)=0$, removing the arbitrary addition of p . Decay supplies the remaining solvability condition and determines e . Independently, multiply the equation by p and integrate by parts over the real line. Since the norm of p squared is exactly $2/3$, the scalar moment is

$$e = \frac{3}{2} \int_{-\infty}^{\infty} p^2 (12uA + 2H + 3u^2 + 1) dx.$$

The implementation obtains e both as a boundary-problem parameter and through this independent Fredholm moment. Their numerical values agree at about fourteen decimal places in the executed calculation. The physical energy expansion, divided by v squared, is $E=3\lambda/2+(2h_0^2/\lambda)e\kappa^2+O(\kappa^3)$. Thus the source background distortion, the virtual Higgs response and the source mode all contribute to the same derived energy coefficient. Keeping the source mode unchanged would miss one of the ingredients needed for the next radiation cancellation.

The third-order open-channel forcing. Let the unmixed Higgs threshold be $m_H^2/v^2=\lambda+C\kappa+Q\kappa^2+O(\kappa^3)$. Expanding the canonical Hessian and mode equation gives a third-order Higgs source consisting of a Higgs-only piece and a piece proportional to h_0^2 . With $u=\tanh(x)$, these universal forcing functions are

$$f_H = 2uTp + (6H + C - s^2)B + s(6T + 3H^2 + 3CH),$$

$$f_S = 2uW + 2Ap - es.$$

The remaining adjustable term is $Q\kappa^2$ times s , after the common factor h_0/k to the fifth is removed. Its cosine overlap is nonzero. Put $F=\pi/\cosh(\pi/2)$, $D_H=-\int \cos(x) f_H(x) dx/F$, and $D_S=-\int \cos(x) f_S(x) dx/F$. The new coefficient is

$$Q = \frac{2}{\lambda} (D_H + h_0^2 D_S).$$

This separation is exact algebra in the perturbative hierarchy. The universal numbers are numerical integrals of its solutions: D_H approximately 0.0541139876765, D_S approximately 2.7147310127655 and e approximately 3.1191218007647. At $\lambda=0.1$ and $h_0=1$, they give Q approximately 55.3769000088 and energy coefficient approximately 62.3824360153. The relation is a derived scalar-wall coefficient, not a fitted nonlinear quartic or a prediction of measured flavor parameters.

Exact polynomial bookkeeping. The new symbolic calculation extracts the static and fluctuation force coefficients directly from the square action as rational symbolic expressions. Fifteen identities reduce exactly to zero. They check the order-one source zeros, both order-two background forces, all relevant mode forces through order three, the universal third-force separation, the exact source wall and shape equations, the localized leading Higgs response and the shape normalization. A particularly consequential identity is that the order-two Higgs diagonal Hessian contains no U^2 source-background term: $\kappa(u^2-1)$ first acquires U^2 at order three, and the Higgs mode itself begins at order one. That term therefore first enters the Higgs-mode equation at order four. The symbolic checks establish these algebraic identities in SymPy; they do not claim a new Lean theorem or certify numerical differential solutions.

Why the formal construction continues. The static source operator is minus d^2/dx^2+4-6s^2 . Its translation zero mode is even, while the source corrections here are odd; the odd inverse is therefore unambiguous. The static Higgs operator minus d^2/dx^2+2 is strictly positive. The source shape operator minus d^2/dx^2+1-6s^2 has one odd kernel, p . At each order the next energy coefficient cancels its p projection, and the fixed central derivative removes the remaining kernel freedom. The even open operator d^2/dx^2+1 admits a decaying solution precisely when its real cosine forcing moment vanishes. If that condition holds, its decaying solution is given by the terminal Green integral of $\sin(t-x)$ times the forcing from x to infinity; the cosine moment imposes the derivative-zero parity condition at the origin.

At the next portal order the new threshold coefficient multiplies the leading localized Higgs response s . Its solvability multiplier is the strictly positive transform $F=\pi/\cosh(\pi/2)$. Consequently one real quartic coefficient removes one real radiation condition. Lower response coefficients determine the other forcing terms. The resulting triangular recursion first determines the next mode radiation cancellation, then the static Higgs response that depends on that already selected quartic coefficient. The source static correction and source energy/mode correction use previously selected Higgs data. All finite-order forcings are smooth, localized combinations of earlier responses. The inverses and moment constraints above therefore specify a unique formal coefficient hierarchy after parity, fixed vacuum and mode normalization have been chosen. The statement concerns formal perturbation coefficients; it supplies neither bounds controlling their growth nor convergence of their sum.

The leakage-order consequence. The scalar-node first induced Higgs response is already localized, so its order-one radiation amplitude vanishes. Leaving the threshold uncorrected permits an order-two radiation amplitude and a fourth-power width. Correcting C removes that amplitude; correcting Q removes the order-three amplitude. A correction through threshold order n therefore formally leaves radiation beginning at order $n+2$ and width beginning at order $2n+4$, provided the first remaining amplitude is nonzero and the outgoing normalization remains regular. This is an asymptotic mechanism derived from the amplitude equation. It does not say that every finite portal has precisely that pure power, or that a formally canceled infinite series defines a stable particle.

Independent nonlinear and outgoing controls. The driver compares predictions with twelve joint background/mode solves across λ, h_0 pairs $(0.1,1)$, $(0.2,0.5)$ and $(0.07,1.5)$. As portal decreases, the quotient $[m_H^2/v^2 - \lambda C]/\lambda^2$ approaches the predicted Q . The corresponding source-energy difference quotient approaches its predicted coefficient. These solves do not supply input values to the response hierarchy. In the unit- v , $\lambda=0.1$, $h_0=1$ action, the quadratic-corrected thresholds give positive outgoing widths approximately 8.31794×10^{-24} , 2.18984×10^{-21} , 5.94480×10^{-19} and 1.70339×10^{-16} GeV at portals 0.000125, 0.00025, 0.0005 and 0.001. Successive doubling ratios are approximately 263.27, 271.47 and 286.53; the smallest-portal ratio is closest to the eighth-power value 256. The corresponding first-order correction retains sixth-power behavior. No lifetime or exactly zero leakage is inferred.

Numerical controls and scope. Response domains $x=24,28,32$ are retained, with all five residuals and quadrature estimates. The default radius 32 controls the small finite-tail error in the open response and its fitted boundary parameter C ; this parameter is determined by a linear forcing solvability equation rather than fitting a radiation width. The finite-radius B response uses zero value and derivative at the endpoint. A uses zero endpoint value; H and T use decaying free Robin conditions. W uses the forced asymptotic condition $W' + W = -(4-e)\pi/2$, preserving the leading $x \exp(-x)$ energy/threshold tail. The last two domain values agree in D_H and D_S within about five times ten to the minus eleven. Such agreement is evidence, not a certified tail bound. Independent outgoing solves use twenty-two scalar decay lengths and tolerance ten to the minus eleven; their probability-current identities remain consistent even at the smallest recorded width. No finite-domain zero is promoted to an infinite-domain theorem.

Reproduction and the next frontier. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_wall_fredholm_hierarchy.py`. The new module is `python/perfectpower/wall_fredholm_hierarchy.py`, with nine focused tests in `python/tests/test_wall_fredholm_hierarchy.py` and a complete scientific receipt under `receipts/flavor_cosmology`. Fifty-four related wall tests also pass in this execution. Plot with `python scripts/plot_wall_fredholm_hierarchy.py` and render with `python scripts/render_wall_fredholm_hierarchy.py`. The standalone account preserves the concurrent arithmetic, commutant, gauge and scalar-flavor work. The expansion is presently for $g=0$; the earlier Gaussian-coupled 154.23 GeV candidate is not assigned these coefficients. Extending the hierarchy to that coupled channel, bounding the formal coefficients, proving an actual radiation zero and testing quantum persistence remain substantive new mathematics and physics.

The broader literature treats preservation of embedded states as a parameter-dependent solvability problem. Amgad Abdrabou, Lijun Yuan, Wangtao Lu and Ya Yan Lu, Parametric dependence of bound states in the continuum: a general theory, arXiv:2205.09951, Phys. Rev. A 107, 033511 (2023), <https://arxiv.org/abs/2205.09951>, provides useful context in photonic structures. Its codimension statements are not imported as a theorem for this scalar action. Our specific contribution is the square-action coefficient hierarchy, its universal second retuning and shape-energy coefficients, the formal parity/moment construction and independent wall/outgoing tests.